



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

4049/01

Paper 1

23 August 2023

2 hours 15 minutes

Marking Scheme

Marking Scheme for 2023 Prelim Add Math Paper 1

Qn	Solution	Marks / Comments
1(a)	$\left[\left(\frac{20}{3} \right)^{-3} \times \sqrt{3^5} \right] \div \frac{3}{2} = 2^a 3^b 5^c$ <i>LHS</i> $\left(\frac{3}{2^2 \times 5} \right)^3 \times 3^{\frac{5}{2}} \times \frac{2}{3}$ $= 2^{-5} 3^{4.5} 5^{-3}$ $a = -5, b = 4.5, c = -3$	B1 (for a , b and c)
1(b)	$2^{2x+1} + 4^{x-1} = 2(3^{1-x})$ $2^{2x+1} + 2^{2x-2} = 2(3^{1-x})$ $2^{2x} + 2^{2x-3} = (3^1)(3^{-x})$ $2^{2x} \left(1 + \frac{1}{8} \right) = \frac{3}{3^x}$ $(4^x)(3^x) = \frac{3}{1\frac{1}{8}}$ $12^x = 2\frac{2}{3} \text{ (shown)}$	M1 (for changing to common base 2) M1 ($2^{2x} (1 + \frac{1}{8})$ or similar) A1
2(a)	$\log_2 x^2 = 3 \log_x 2 + 1$ $2 \log_2 x = \frac{3 \log_2 2}{\log_2 x} + 1$ <i>Let $\log_2 x = m$</i> $2m = \frac{3}{m} + 1$ $2m^2 = 3 + m$ $2m^2 - m - 3 = 0$ $m = 1.5, -1$ $\log_2 x = 1.5 \quad \text{or} \quad \log_2 x = -1$ $x = 2.83 \quad \quad \quad x = 0.5$	M1(change base) M1 (forming equation or equivalence) M1 (find m or equivalent) A1 (method shown)

2(b)	$\ln a - 2 \ln b = \ln(a + b)$ $\ln\left(\frac{a}{b^2}\right) = \ln(a + b)$ $\frac{a}{b^2} = a + b$ $a = ab^2 + b^3$ $a - ab^2 = b^3$ $a = \frac{b^3}{1 - b^2}$	M1 ($\ln b^2$) M1 (form $\ln\left(\frac{a}{b^2}\right)$) A1
3(a)	$(\sqrt{5} + 1)x = \sqrt{3}$ $x = \frac{\sqrt{3}}{\sqrt{5} + 1}$ $= \frac{(\sqrt{3})(\sqrt{5} - 1)}{(\sqrt{5})^2 - 1^2}$ $= \frac{\sqrt{15} - \sqrt{3}}{4}$ $x + \frac{1}{x} = \frac{\sqrt{15} - \sqrt{3}}{4} + \frac{4}{\sqrt{15} - \sqrt{3}}$ $= \frac{\sqrt{15} - \sqrt{3}}{4} + \frac{4(\sqrt{15} + \sqrt{3})}{12}$ $= \frac{7\sqrt{15} + \sqrt{3}}{12}$ $= \frac{7}{12}\sqrt{15} + \frac{1}{12}\sqrt{3}$	M1(rationalisation) M1 (simplification) A1(for both correct)
3(b)	$2\sqrt{2} - 3 = \frac{\sqrt{h - k\sqrt{2}}}{1 + \sqrt{2}}$ $(2\sqrt{2} - 3)(1 + \sqrt{2}) = \sqrt{h - k\sqrt{2}}$ $1 - \sqrt{2} = \sqrt{h - k\sqrt{2}}$ $1 - 2\sqrt{2} + 2 = h - k\sqrt{2}$ $3 - 2\sqrt{2} = h - k\sqrt{2}$ $h = 3, k = 2$	M1($(2\sqrt{2} + 4 - 3 - 3\sqrt{2})$ or better) M1 (for expansion $(1 - \sqrt{2})^2$) A1 (both answer correct)

5(a)	$\frac{11x^2 - 8x + 3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$ $11x^2 - 8x + 3 = A(x^2+1) + (Bx+C)(x-1)$ $\text{let } x=1,$ $11-8+3=2A$ $A=3.$ $\text{let } x=0,$ $3=A+C$ $C=0.$ $\text{Compare } x^2,$ $B=8.$ $\frac{11x^2 - 8x + 3}{(x-1)(x^2+1)} = \frac{3}{x-1} + \frac{8x}{x^2+1}$	M1 (recall correct partial fraction) A1 A1 A1 [-1] if not in partial fraction form
5(b)	$\int \frac{11x^2 - 8x + 3}{(x-1)(x^2+1)} dx$ $= \int \frac{3}{x-1} + \frac{8x}{x^2+1} dx$ $= 3\ln(x-1) + 4\ln(x^2+1) + c$	M2 (M1 for $3\ln(x-1)$ and $4\ln(x^2+1)$) A1 (-1 mark if no c)

6	$f'(x) = \sin 3x - \cos \frac{x}{2}$ $f''(x) = 3 \cos 3x + \frac{1}{2} \sin \frac{x}{2}$ $f(x) = \int \sin 3x - \cos \frac{x}{2} dx$ $= -\frac{1}{3} \cos 3x - 2 \sin \frac{x}{2} + c$ <i>Sub</i> $x = \frac{\pi}{3},$ $0 = -\frac{1}{3} \cos \pi - 2 \sin \frac{\pi}{6} + c$ $c = \frac{2}{3}.$ $f(x) = -\frac{1}{3} \cos 3x - 2 \sin \frac{x}{2} + \frac{2}{3}$ <i>L.H.S</i> $12(3 \cos 3x + \frac{1}{2} \sin \frac{x}{2}) + 3(-\frac{1}{3} \cos 3x - 2 \sin \frac{x}{2} + \frac{2}{3})$ $= 36 \cos 3x + 6 \sin \frac{x}{2} - \cos 3x - 6 \sin \frac{x}{2} + 2$ $= 35 \cos 3x + 2 = R.H.S (shown)$	M2(M1 for each differentiation) M2(M1 for each differentiation) M1(value of c) M1 (correct expansion) A1
7(a)	$(1 + y - y^2)^n = 1 + {}^nC_1(1)(y - y^2) + {}^nC_2(1)(y - y^2)^2 + \dots$ $= 1 + ny - ny^2 + \frac{n(n-1)}{2}(y^2 - 2y^3 + \dots) + \dots$ $= 1 + ny + (\frac{n(n-1)}{2} - n)y^2 + \dots$ <i>By comparing,</i> $\frac{n(n-1)}{2} - n = 20$ $n(n-1) - 2n = 40$ $n^2 - 3n - 40 = 0$ $n = 8 \text{ or } -5(NA)$ $a = n$ $= 8.$	M1(correct change of nC_r) M1 (attempt to compare y^2) A1 A1

8(c)	<i>Eqn of AD,</i> $y = -\frac{3}{2}x + 4$ <i>Sub $M(k+1, k)$ to AD,</i> $k = -1.5(k+1) + 4$ $2.5k = 2.5$ $k = 1.$ <i>Point $M = (2, 1)$</i> <i>Midpoint of $CF = (2, 1)$</i> $\left(\frac{x+5}{2}, \frac{y+3}{2}\right) = (2, 1)$ $x = -1,$ $y = -1.$ <i>Point $F = (-1, -1)$</i>	B1 (or other method) A1
9(a)	$kx^2 + 4x + 2k = x + k$ $kx^2 + 3x + k = 0$ $D < 0$ $3^2 - 4(k)(k) < 0$ $0 < 4k^2 - 9$ $\text{let } 4k^2 - 9 = 0$ $k = \pm 1.5$ $k < -1.5 \text{ or } k > 1.5 (NA)$	M1(simult. Eqn) M1 (use of $b^2 - 4ac$) M1 (values of k) A1 [-1 mark if never reject 1.5]
9(b)	$y = (m-5)x^2 + 2mx + (m+3)$ $D = (2m)^2 - 4(m-5)(m+3)$ $= 4m^2 - 4(m^2 - 2m - 15)$ $= 8m + 60$ <i>For always positive,</i> $8m + 60 < 0$ $m < -7.5$ However is $m < -7.5$, then $(m-5)$ will be negative (downward curve) and thus will never be always positive.	M1 (use of Discriminant) M1 ($D < 0$) A1 (explanation)

10(a)	$\frac{d}{dx} \left(\frac{\ln x}{4x} \right) = \frac{(4x) \left(\frac{1}{x} \right) - (\ln x)(4)}{(4x)^2}$ $= \frac{4 - 4 \ln x}{16x^2}$ $= \frac{1 - \ln x}{4x^2} \text{ (shown)}$	M1 (numerator) M1 (denominator) A1
10(b)	$\int \frac{d}{dx} \left(\frac{\ln x}{4x} \right) = \int \frac{1 - \ln x}{4x^2} dx$ $\frac{\ln x}{4x} = \int \frac{1}{4x^2} dx - \int \frac{\ln x}{4x^2} dx$ $\frac{1}{4} \int \frac{\ln x}{x^2} dx = \int \frac{1}{4x^2} dx - \frac{\ln x}{4x}$ $\frac{1}{4} \int \frac{\ln x}{x^2} dx = -\frac{1}{4} (x^{-1}) + c - \frac{\ln x}{4x}$ $\int \frac{\ln x}{x^2} dx = -\frac{1}{x} - \frac{\ln x}{x} + c$	M1 (introduce integration) B1 (correct $(-\frac{1}{4} x^{-1})$) A1
11(a)	Let angle NMP = a° Angle MBP = a° (alternate segment theorem) Since Angle NMP = angle MBP = a° Triangle PMB is an isosceles triangle. Angle MRP = Angle MBP = a° (angles in same segment) Angle BRP = Angle BMP = a° (angles in same segment) Angle BRP = Angle MRP Hence RP bisects angle MRB. (shown)	M1 M1 A1 (-1 mark if no explanation given)
11(b)	Angle MRT = Angle PRB (angle bisector) Angle RMT = Angle RPM (angles in same segment) By “angle-angle-angle” property, triangle RTM is similar to triangle RBP.	M1 M1 A1 (-1 mark if no explanation given)

12(a)	$\sin 3x = \sin x(4\cos^2 x - 1)$ <i>L.H.S</i> $\sin 3x = \sin(2x + x)$ $= \sin 2x \cos x + \cos 2x \sin x$ $= 2\sin x \cos^2 x + (2\cos^2 x - 1)\sin x$ $= \sin x(2\cos^2 x + 2\cos^2 x - 1)$ $= \sin x(4\cos^2 x - 1) = R.H.S (shown)$	M1 (or any use of relevant trigo formulas) M1(convert sin2x or cos2x) A1
12(b)	$3\sin 3x = 16\cos x \sin x$ $3\sin x(4\cos^2 x - 1) = 16\cos x \sin x$ $3\sin x(4\cos^2 x - 1) - 16\cos x \sin x = 0$ $\sin x(12\cos^2 x - 3 - 16\cos x) = 0$ $\sin x = 0 \quad \text{or} \quad 12\cos^2 x - 16\cos x - 3 = 0$ $x = 0, \pi \text{ or } 2\pi.$ or $12\cos^2 x - 16\cos x - 3 = 0$ $\cos x = 1.5(NA) \quad \text{or} \quad \cos x = -\frac{1}{6}$ $\text{ref}(x) = 1.4033$ $x = 1.74, 4.54$ <i>Ans : x = 0, 1.74, π, 4.54, 2π.</i>	M1(taking out common factor sin x or expansion) M1 (sin x =0) A1 (all 3 answers) M1(for trigo equation) A1 (both answers)